Hardware Architecture Design for Iterative Reconstruction Algorithms Toward Palm-Size Photoacoustic Tomography

Yuwei Zheng, Zijian Gao, Yuting Shen, Jiadong Zhang, Ruixi Sun, Daohuai Jiang, Fengyu Liu, Xiran Cai, *Member, IEEE*, Feng Gao, Yuan Gao, *Member, IEEE*, and Fei Gao, *Member, IEEE*

***Abstract*—Photoacoustic (PA) imaging technology integrates the deep penetration depth of ultrasound imaging with the high resolution of optical imaging, demonstrating significant potential in biomedical applications. Many preclinical studies and clinical applications urgently require a portable, high-quality, low-cost, and fast imaging system. Thus, translating advanced image reconstruction algorithms into hardware implementations is highly desired. However, existing iterative PA image reconstructions, although exhibit higher accuracy than the delay-and-sum algorithm, suffer from high computational cost. In this paper, we introduce a novel model-based hardware architecture for palm-size PA tomography (palm-PAT), aiming at enhancing both the speed and performance of image reconstruction at a much lower system cost. To achieve this, we propose an innovative data reuse method that significantly reduces the consumption of hardware storage resources, achieving a reduction of approximately 75% in hardware storage requirements. We conducted experiments utilizing the FPGA implementation of the algorithm, using phantom, human finger data in vivo and ex vivo breast tumor data to verify the feasibility of the proposed method. The results demonstrate that our proposed architecture can substantially reduce system cost while maintaining high imaging performance. The novel hardware architecture design of the model-based algorithm achieves a speedup of up to approximately 270 times compared to the CPU, while the corresponding energy efficiency ratio is improved by more than 2700 times.**

***Index Terms*—Photoacoustic imaging, image reconstruction, model-based, hardware architecture, palm-size, FPGA.**

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Yuwei Zheng, Ruixi Sun, Fengyu Liu, and Xiran Cai are with the School of Information Science and Technology, ShanghaiTech University, Shanghai 201210, China (e-mail: zhengyw2022@shanghaitech.edu.cn; sunrx2023@shanghaitech.edu.cn; liufy@shanghaitech.edu.cn; caixr@shanghaitech.edu.cn).

Zijian Gao was with the School of Information Science and Technology, ShanghaiTech University, Shanghai 201210, China. He is now with the Faculty of Medicine, The Chinese University of Hong Kong, Hong Kong, China (e-mail: zijian.gao@link.cuhk.edu.hk).

Yuting Shen was with the School of Information Science and Technology, ShanghaiTech University, Shanghai 201210, China. She is now with United Imaging Healthcare Technology Company Ltd., Shanghai 201210, China (e-mail: shenyt@shanghaitech.edu.cn).

Jiadong Zhang was with the School of Information Science and Technology, ShanghaiTech University, Shanghai 201210, China. He is now with the Department of Biomedical Engineering, City University of Hong Kong, Hong Kong, China (e-mail: jiadzhang3-c@my.cityu.edu.hk).

Daohuai Jiang was with the School of Information Science and Technology, ShanghaiTech University, Shanghai 201210, China. He is now with the College of Photonic and Electronic Engineering, Fujian Normal University, Fuzhou 350007, China (e-mail: jiangdh@fjnu.edu.cn).

Feng Gao was with the School of Information Science and Technology, ShanghaiTech University, Shanghai 201210, China. He is now with Innosilicon Technology Company Ltd., Xi'an 710000, China (e-mail: gaofeng@shanghaitech.edu.cn).

Yuan Gao is with the School of Microelectronics, Southern University of Science and Technology, Shenzhen 518055, China (e-mail: gaoy@sustech.edu.cn).

Fei Gao is with the School of Biomedical Engineering, Division of Life Sciences and Medicine, and the School of Engineering Science, University of Science and Technology of China, Hefei, Anhui 230026, China, and also with the Suzhou Institute for Advanced Research, University of Science and Technology of China, Suzhou, Jiangsu 215123, China (e-mail: fgao@ustc.edu.cn).

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I. Introduction

PHOTOACOUSTIC (PA) imaging (PAI) is an emerging hybrid medical imaging modality that leverages the photoacoustic effect, where optical absorption by biological tissues induces thermoelastic expansion, subsequently generating ultrasound waves [1], [2], [3]. This technique uniquely combines the high optical contrast with the deep tissue penetration capabilities of ultrasound [4], [5], [6], offering promising applications in both pre-clinical research and clinical settings. The increasing demand for portable devices in outdoor or emergency scenarios necessitates rapid and high-quality imaging solutions, posing stringent requirements on the cost, miniaturization, and performance of PA imaging systems.

Among various PA image reconstruction algorithms, the delay-and-sum (DAS) beamforming algorithm is widely adopted due to its simplicity and low computational cost. However, DAS is prone to severe artifacts, especially under limited-view or sparse sampling conditions. To address these limitations, several variants of DAS, such as delay-multiply-and-sum (DMAS) [7], [8] and delay-and-sum with coherence factor (DAS-CF) [9], have been developed and accelerated using field-programmable gate arrays (FPGAs), thereby enhancing reconstruction speed [10]. Another commonly used one step method is time-reversal reconstruction [11], which, despite its efficiency, still suffers from critical artifacts.

Currently, FPGA-accelerated reconstruction and signal processing techniques predominantly focus on simple DAS-based methods [10], [11], [12], [13] and back projection (BP) algorithm [14]. While these methods offer computational efficiency, they often fall short of delivering high image

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quality. In contrast, iterative model-based reconstruction algorithms [15] have demonstrated superior image quality. Previous model-based reconstruction methods employ diverse forward models and regularization strategies, each with inherent trade-offs. For example, truncated singular value decomposition of a frequency-domain model matrix obtained by solving the discretized wave equation accelerates inversion compared to time-reversal while preserving image quality [16], but discards high-frequency details and requires recomputing the model for varying tissue properties. Tikhonov-based least-squares solvers using blur matrices can mitigate artifacts but incur computational overhead in estimating the system resolution [17]. More recently, Learned Fourier neural operators replace the numerical wave solver as the forward operator and speed up reconstruction [18], yet demand large training datasets and may introduce subtle high-frequency errors. Compensation for spatially varying sound speed and optical fluence via Monte Carlo-based iterations [19] improves accuracy in heterogeneous media but relies on precise optical and acoustic priors, which is computationally intensive. Hybrid physics–deep-learning frameworks integrate the wave equation with convolutional networks to refine absorption estimates but require extensive simulations and high-fidelity fluence models [20]. Likewise, diffusion model-based frameworks alternately perform generative steps and data consistency updates via gradient descent to capture complex structures and mitigate oversmoothing [21], but they typically need hundreds of iterations even on GPUs to converge. Although GPU [22] can accelerate these iterative methods [23], [24], including compressed sensing [25] and deep learning frameworks [26], [27], [28], [29], [30], [31], [32], they do not satisfy the low-cost, low-power consumption, and miniaturization requirements essential for portable PA imaging devices. Therefore, there is a pressing need to accelerate these iterative algorithms cost-effectively.

In this paper, we present a novel hardware architecture based on a model-based iterative algorithm [33], [34] that employs the superposed wave (s-Wave) technique [35], [36] to speed up PA image reconstruction while maintaining high image quality. Our approach introduces a novel data reuse method leveraging the geometric symmetry of ultrasound transducer distributions, significantly reducing on-chip storage requirements. The proposed design utilizes DAS as the backward model to transform signals into images, while the s-Wave serves as the forward model to convert images back into signals, simplifying the process of solving complex numerical equations or evaluating a learned neural network into basic multiplication and summation operations, thereby reducing algorithmic complexity and runtime. These operations are implemented using multi-channel parallelism and a pipelining scheme to accelerate data processing. We conducted PA imaging of phantom, human finger in vivo, breast tumor on FPGA platforms to demonstrate the feasibility and potential applications of our proposed approach, indicating its promise for integration into palm-PAT system.

II. Methods

A. Photoacoustic Imaging Principle

PAI aims to recover the initial pressure distribution generated by light absorption, through image reconstruction from the detected PA signals. In a non-absorbing, acoustically homogeneous medium, PA wave propagation obeys below equation:

$$\left(\nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\right) p(\mathbf{r},t) = -\frac{\beta}{\kappa c^2}\frac{\partial^2 T(\mathbf{r},t)}{\partial t^2} \tag{1}$$

where c is the speed of sound, $p(\mathbf{r},t)$ is the acoustic pressure at position $\mathbf{r}$ and time t, β is the thermal expansion coefficient, κ is the isothermal compressibility, and $T(\mathbf{r},t)$ is the temperature rise. Under thermal confinement, heat diffusion is given by:

$$\rho C_V \frac{\partial T(\mathbf{r},t)}{\partial t} = H(\mathbf{r},t) \tag{2}$$

where $H(\mathbf{r},t)$ is the heating function, which defines the thermal energy delivered per unit volume per unit time. By combining (1) and (2), the PA wave equation can be obtained:

$$\left(\nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\right) p(\mathbf{r},t) = -\frac{\beta}{C_P}\frac{\partial H(\mathbf{r},t)}{\partial t} \tag{3}$$

where C_P is the specific heat capacity at constant pressure. The left-hand side of (3) describes wave propagation, while the right-hand side is the time derivative of the heating function $H(\mathbf{r},t)$, which serves as the source term, implying that acoustic waves arise only from temporally varying heating. In PAI, detectors are typically arranged on an enclosing surface S_{sf}. Ultrasound transducers are typically placed around the object on an enclosing surface S_{sf} in spherical, cylindrical, or planar geometries. Given measured surface pressures $p_d(\mathbf{r}_d,t)$ for $\mathbf{r}_d \in S_{sf}$, $t>0$, the initial pressure distribution $p_0(\mathbf{r})$ must satisfy:

$$\nabla^2 p(\mathbf{r},t) - \frac{1}{c^2}\frac{\partial^2}{\partial t^2} p(\mathbf{r},t) = 0, r \in \mathbb{R}^3, t \geq 0 \tag{4}$$

$$p(\mathbf{r}, t=0) = p_0(\mathbf{r}) \tag{5}$$

$$\frac{\partial}{\partial t} p(\mathbf{r}, t=0) = 0 \tag{6}$$

$$p(\mathbf{r}_d,t) = p_d(\mathbf{r}_d,t), \mathbf{r}_d \in S_{sf}, t>0 \tag{7}$$

Solving this set of partial differential equations requires using the free-space Green's function $G(\mathbf{r},\mathbf{r}',t,t')$, which relates the pressure at two distinct space-time points:

$$G(\mathbf{r},\mathbf{r}',t,t') = \frac{\delta((t-t') - |\mathbf{r}-\mathbf{r}'|/c)}{|\mathbf{r}-\mathbf{r}'|} \tag{8}$$

Therefore, the solution to the PA wave equation can be expressed as:

$$p(\mathbf{r}_d,t) = \frac{\beta}{4\pi C_P}\frac{\partial}{\partial t}\int d\mathbf{r}' \frac{1}{|\mathbf{r}_d-\mathbf{r}'|} H\left(\mathbf{r}', t - \frac{|\mathbf{r}_d-\mathbf{r}'|}{c}\right) \tag{9}$$

Under the acoustic stress-confinement condition, a short light pulse allows us to approximate heating as, $H(\mathbf{r}',t') \approx A(\mathbf{r}')\delta(t')$, where $A(\mathbf{r}')$ is the absorbed energy density at $\mathbf{r}'$. Substituting this into the above equation yields:

$$p(\mathbf{r}_d,t) = \frac{\beta}{4\pi c^2}\frac{\partial}{\partial t}\left[\frac{1}{ct}\int d\mathbf{r}' p_0(\mathbf{r}')\delta\left(t - \frac{|\mathbf{r}_d-\mathbf{r}'|}{c}\right)\right] \tag{10}$$

B. Paml-PAT

Our envisioned palm-PAT system, as shown in Fig. 1(a), is designed to be entirely handheld. The front end of the system features ultrasound transducers for receiving PA signals, with an adjacent LED [37], [38], which is able to operate at high repetition rates of several kHz [39] emitting pulsed laser light. The entire PA imaging hardware is housed within the device

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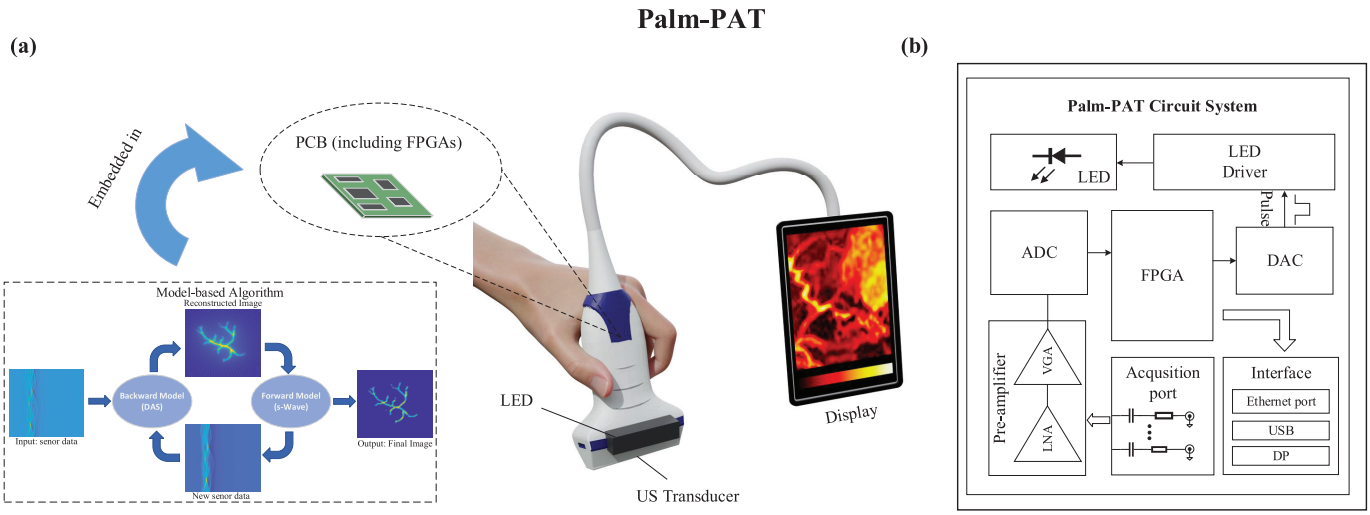


Fig. 1. An envisioned Palm-PAT system. It integrates the ultrasound transducer, LED, data acquisition module, image reconstruction hardware module, and high-speed data interfaces. (a) Diagram of the palm-PAT. PCB: printed circuit board, LED: light emitting diode, US Transducers: Ultrasound Transducers. (b) Palm-PAT circuit system. ADC: analog to digital converter, DAC: digital to analog converter, LNA: low noise amplifier, VGA: variable gain amplifier, USB: universal serial bus, DP: DisplayPort.

body and integrated onto printed circuit boards (PCB). A noteworthy aspect is the implementation of various PA image reconstruction algorithms (e.g., DAS, DMAS, Model-based) on FPGA (or application-specific integrated circuits (ASIC) chip in future work) and different algorithms can be rapidly and seamlessly interchanged through partial reconfiguration of the FPGA which is integrated into the hardware system, enabling low-latency and real-time image reconstruction. The system framework of the palm-PAT is illustrated in Fig. 1(b). The front end includes a preamplifier, filter, and ADC for signal amplification, filtering, and analog-to-digital conversion. The FPGA is responsible for generating the laser diode drive signal, controlling the entire data acquisition process, performing image reconstruction, and managing data transmission. The interface section includes ports such as DP, USB, and ethernet and so on, facilitating connections to external devices like displays and personal computers (PCs) for real-time image display and high-speed data transmission.

Integrating the entire reconstruction pipeline, including forward and backward models, iterative updates, and thresholding, into a single FPGA chip, yields substantial reductions in size, weight, and power consumption compared to GPU or CPU workstations, enabling the PCB-integrated palm-PAT device to be truly handheld and even battery-operated. This miniaturization allows the probe to be placed directly on the skin or incorporated into endoscopic or catheter-based platforms for intraoperative, bedside, or field applications, while its low-latency, fast output delivers immediate visual feedback of vascular structures or tumor margins, facilitating rapid clinical decisions without bulky equipment, reducing capital and maintenance costs, and simplifying sterilization and power/thermal management in operating rooms. Ultimately, this approach broadens access to high-contrast, point-of-care photoacoustic imaging in outpatient clinics, emergency departments, and remote settings.

C. Algorithms Design

1) Backward Model: Delay-and-Sum Algorithm: Delay-and-Sum (DAS) is the most commonly used method [10] for rapid PA image reconstruction. The specific description is shown in Algorithm 1. It processes the input sensor data based on the spatial relationship between each ultrasound transducer element (sensor) and each pixel position. Before performing the reconstruction, the positions of the sensors are fixed so that we can get the distances between the sensors and various positions within the imaging region in advance. Therefore, the delay in DAS can be expressed as

Algorithm 1 Delay-and-Sum

Require: Input sensor data S; the number of sensors N; ROI size $n \times m$; pixel position (x_p, y_p); sensors position (x_s, y_s); Speed of sound c; the sample frequency f_s.
Ensure: Reconstructed image S_{DAS}.

$I_{DAS} = 0$
for $i \leftarrow 1, N$ **do**
  for $j \leftarrow 1, n \times m$ **do**
    $d_{ij} = \sqrt{(x_{pj} - x_{si})^2 + (y_{pj} - y_{si})^2}$
    $delay(x_{pj}, y_{pj}, i) = \frac{d_{ij}}{c} \cdot f_s$
    $I_{DAS}(j) = I_{DAS}(j) + S\left(i, delay\left(x_{pj}, y_{pj}, i\right)\right)$
  end for
end for
return Reconstructed image I_{DAS}

$$delay(x, y, i) = \frac{d(x, y, i)}{c} \cdot f_s. \tag{11}$$

where $d(x, y, i)$ represents the distance from position (x, y) to the ith sensor. c is the speed of sound(1500 m/s), and f_s is the sampling frequency. By extracting and summing the collected sensor data based on these delay values, we can obtain the final image values $I_{DAS}(x, y)$, which can be expressed as

$$I_{DAS}(x, y) = \sum_{i=1}^{N} S\left(i, delay\left(x, y, i\right)\right) \tag{12}$$

where $I_{DAS}(x, y)$ represents the pixel value at position (x, y) within the imaging region. N denotes the number of sensors.

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$S\left(i, delay\left(x, y, i\right)\right)$ represents the data value in the sensor data received by the ith transducer element with a delay of $delay(x, y, i)$. After traversing the above calculations over the entire imaging plane, we can obtain the reconstructed image.

2) Forward Model s-Wave Algorithm: In contrast to DAS, which performs back-projection from transducer-acquired sensor data to reconstruct a PA image, the superposed Wave (s-Wave) algorithm is a forward simulation method to simulate the acoustic signals that would be detected at the sensors from a given PA image. Instead of solving the full wave equation numerically in MATLAB k-Wave toolbox [40], s-Wave approximates the forward propagation process using a combination of amplitude scaling and phase delay applied to a precomputed standard waveform, significantly improving simulation speed while ensuring that the results closely resemble the signals from k-Wave [34]. We found that each pixel value in a PA image generates a set of signals after the forward process, and the shapes of the signals obtained at different pixels are very similar, mainly differing only in magnitude and phase. The differences can be characterized by the spatial relationship between pixels and sensors.

Algorithm 2 s-Wave

Require: Input PA image I; Standard sensor data s; the reference distance d_s; the number of sensors N; ROI size $n \times m$; pixel position (x_p, y_p); sensors position (x_s, y_s); Speed of sound c; the sample frequency f_s; constant k.
Ensure: New sensor data s_n.

for $i \leftarrow 1, N$ **do**
  for $j \leftarrow 1, n \times m$ **do**
    $d_{ij} = \sqrt{\left(x_{pj} - x_{si}\right)^2 + \left(y_{pj} - y_{si}\right)^2}$
    $\tau\left(d_{ij}, d_s\right) = \frac{d_{ij} - d_s}{f_s} \cdot f_s$
    $A\left(d_{ij}, I_j\right) = k\frac{f_j}{d_{ij}^2}$
    $s_{ij} = A\left(d_{ij}; I_j\right) \cdot \left\langle s|\tau\left(d_{ij}, d_s\right)\right\rangle$
  end for
  $s_n^{(i)} = \sum_{j=1}^{n \times m} s_{ij}$
end for
$s_n = \sum_{i=1}^{N} s_n^{(i)}$
return Reconstructed image s_n

Initially, s-Wave utilizes k-Wave to obtain a set of standard sensor data, denoted as $s \in \mathbb{R}^{1 \times M}$ (M is the sampling depth), which is acquired from unit pixels at the center of the imaging region. Then, the phase will change across the entire time domain range of the signal based on the offset as expressed as

$$\tau\left(d_p, d_s\right) = \frac{d_p - d_s}{c} \cdot f_s \tag{13}$$

where $\tau\left(d_p, d_s\right)$ represents the time delay difference between the other pixel and the central pixel position. The d_p denotes the distance from the specified pixel to each sensor, and d_s is the distance from the central pixel to each sensor. There are three factors that will influence the amplitude of sensor data: pixel value, energy dissipation and acoustic attenuation. To simplify the process, we ignore the influence of attenuation in this work [41]. Furthermore, because the amplitude of the sensor data is proportional to the pixel value p, we adjust the signal amplitude simply by multiplying the sensor data by p. To account for energy dissipation during acoustic propagation [42], [43], we model this attenuation using the reciprocal of a power-law function. The amplitude coefficient is related to the intensity of pixel values and the energy attenuation in the medium so that the coefficient can be defined as

$$A\left(d_p; p\right) = k\frac{p}{{d_p}^2} \tag{14}$$

where $A\left(d_p; p\right)$ is the amplitude coefficient for the specified pixel position. Here, k is a constant, and p represents the pixel value. Therefore, the signal $s_p \in \mathbb{R}^{1 \times M}$ obtained at the specified pixel point can be expressed as

$$s_p = A\left(d_p; p\right) \cdot \left\langle s|\tau\left(d_p, d_s\right)\right\rangle \tag{15}$$

this equation simulates the physical process of how a pixel with intensity p contributes to the signal received at a specific sensor location, where $\left\langle s|\tau\left(d_p, d_s\right)\right\rangle$ models the phase shift in time caused by the different propagation delays, applied to a precomputed standard waveform s. When $\tau > 0$, it means $d_p > d_s$, and we have to right-shift s to get the right phase. Otherwise, we have to left-shift s. In physical terms, this formulation allows us to simulate, with high efficiency, what the sensor would detect if a pressure source of magnitude p were located at a given pixel. It forms the core of our forward model (s-Wave), used in the iterative reconstruction framework to map from image space back to synthetic sensor data, enabling comparison with measured data and driving reconstruction updates. By traversing all pixels, we can get many groups of signals based on s. The number of groups is the same as the number of pixels. By superimposing these signals together, we can get a new set of signals $s_{np} \in \mathbb{R}^{1 \times M}$. Then we traverse all the sensors and repeat the above operation to obtain the new PA signal $s_n \in \mathbb{R}^{N \times M}$ (N is the number of sensors). The specific description is shown in Algorithm 2.

Algorithm 3 Model-Based Reconstruction With s-Wave

Require: Input sensor data S; max iteration number K; initiallearning rate lr; loss threshold L.
Ensure: Final reconstructed image I_t.

$I_0 = DAS(S)$ ▷ Get the initial reconstructed image
$s_n = sWave(abs(I_0))$ ▷ Get the new sensor data
$R_1 = S - s_n$ ▷ Compute the difference
for $t \leftarrow 1, K$ **do** ▷ Iterate and get the final image
  $I'_t = DAS(R_t)$
  $I_t = I_{t-1} + lr * I'_t$ ▷ Weighted superposition
  $R_{t+1} = S - sWave(abs(I_t))$ ▷ residual sensor data
  if $\|R_{t+1}\|_2 < L$ **then** ▷ threshold judgment
    return
  end if
end for
return Reconstructed image I_t

3) Model-Based PA Image Reconstruction Algorithm: The model-based algorithm is an iterative image reconstruction method, in which DAS serves as the backward model from signals to images, and s-Wave acts as the forward model from images to signals [33], [34]. The algorithm's fundamental mechanism is to iteratively alternate forward and backward mappings, thereby progressively minimizing the signal-domain loss function until convergence is achieved. The specific description is shown in Algorithm 3.

Initially, the algorithm processes the raw sensor data $S \in \mathbb{R}^{N \times M}$ including all the sensors (total number of sensing

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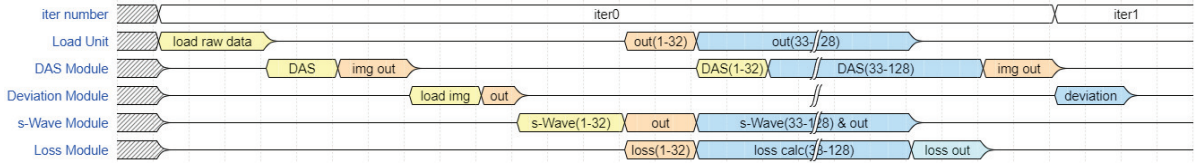


Fig. 2. Timing graph for iteration cycle 0.

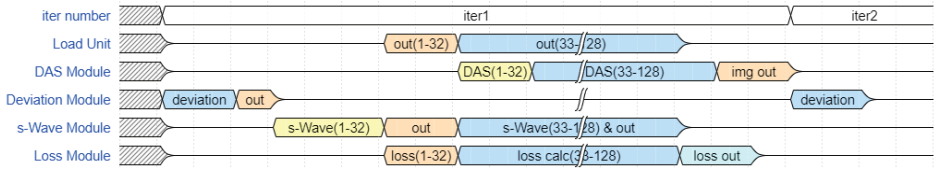


Fig. 3. Timing graph for iteration cycle 1.

elements of the transducer) data using the DAS method to produce an initial reconstructed PA image I_0. This image I_0 is then converted to its absolute value and fed into the s-Wave, which generates new sensor data s_n. The difference between the original sensor data S and the new sensor data s_n results in the residual sensor data R_1. Subsequently, the DAS uses R_1 to create a residual image, which is scaled by a learning rate lr and added to the previously reconstructed image from the DAS. The lr is empirically determined, through extensive experimentation, we found that it must lie within the interval (0-1) to ensure convergence. The lr is set to 0.05 initially and decreases by 0.01 every five iterations until it reaches 0.01. The combined image is then reintroduced into the s-Wave, where its output is once again differenced with the original sensor data S to yield the updated residual data R_{t+1}. During each iteration, a loss value is computed based on R_{t+1}. If this loss falls below a predetermined threshold, the iterative process terminates, and the most recent combined image will be output as the final result. Otherwise, iterations continue until the maximum number of iterations, predefined at the start, is reached.

This iterative approach overcomes the inherent limitation of DAS, which back-projects every signal into all possible image locations and therefore introduces irrelevant information that manifests as artifacts. R_1 denotes the redundant signal component introduced by the DAS algorithm relative to the raw sensor data, which we aim to remove. Therefore, back-projecting R_1 with DAS produces a residual image that captures the majority of the redundant information. Because DAS cannot perfectly represent all redundant components, we apply a weighting factor to this residual image before subtracting it from the image obtained in the previous iteration. Iteration ensures that each subsequent image reconstruction refines the previous one, enhancing the accuracy and quality of the final PA image. By dynamically adjusting and converging towards the optimal image reconstruction, this method leverages both the DAS and s-Wave algorithms in a complementary manner, providing a robust solution for real-time, FPGA-accelerated PAT system.

D. Hardware Architecture

1) Architecture Overview: The hardware acceleration architecture of model-based algorithm consists of 6 sub-modules, namely, the LU, DAS module, Deviation module, s-Wave module, Loss module, and Top Controller module, each represented in different colors as shown in Fig. 4. In the system, the sensor data is ingested as the input and processed to produce the reconstructed image as the output. The LU is tasked with buffering the incoming raw sensor data and ensuring its orderly storage. The DAS module then processes this data by employing the delay-and-sum algorithm to effectively extract and aggregate the sensor data into a corresponding PA image, which is subsequently output. The Deviation module focuses on calculating the difference between successive iterations to generate a residual image, and is also responsible for outputting the final image upon completion of the iteration number. Meanwhile, the s-Wave module performs the forward transformation of the image into sensor data and passes it to the Loss module, where the sensor data residuals and the final loss value are computed. The Top Controller module controls the behavior of each sub-module according to the execution status of the system.

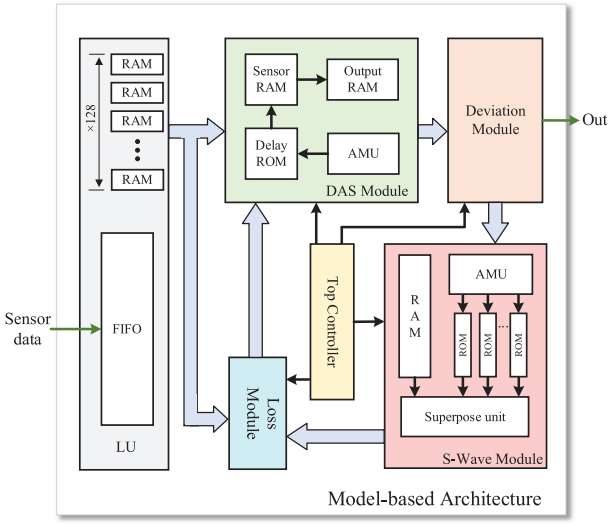


Fig. 4. Hardware architecture of model-based algorithm with s-Wave. FIFO: first in first out; LU: load unit; AMU: address mapping unit.

The timing flow (Fig. 2, Fig. 3) begins with iteration cycle 0, where the LU loads raw data and then the DAS module performs the initial reconstruction. The Deviation module, at this early stage, immediately stored and forwarded DAS

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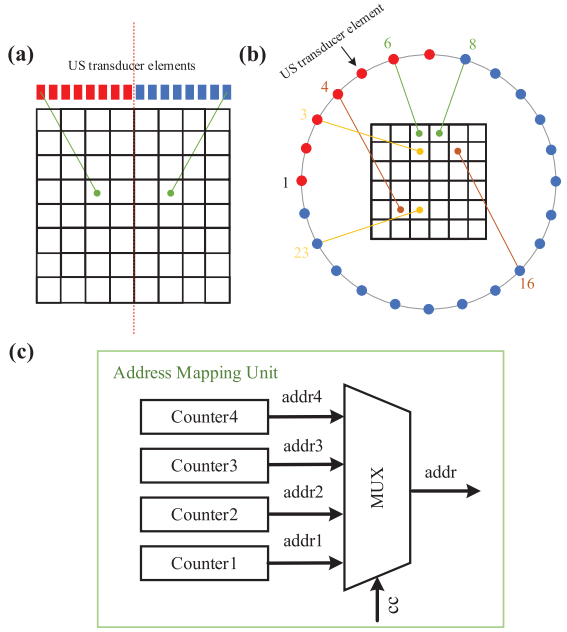


Fig. 5. Proposed data reuse method. (a) Symmetry description of linear transducer array and imaging region. (b) Symmetry description of ring-shaped transducer array and imaging region (c) Hardware architecture of address mapping unit (AMU). cc: execution cycle count.

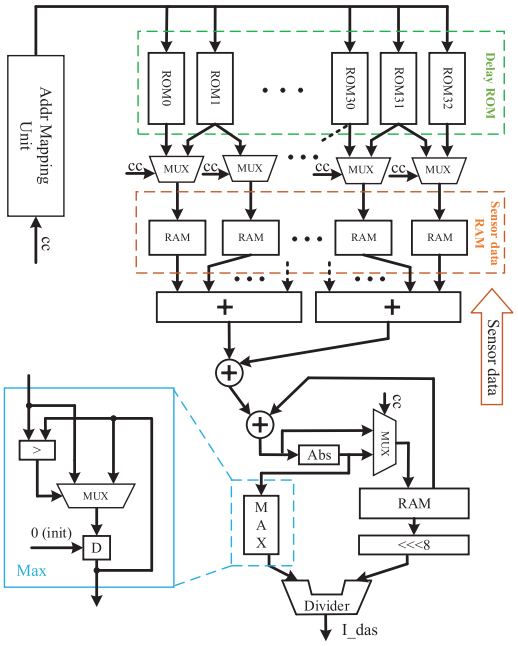


Fig. 6. Structure of the DAS module. D: register. cc: execution cycle count.

output. Next, the s-Wave module performs 32-channel forward projections. Each 32-channel output is paired with its raw data form LU in the Loss module to compute residuals, which feed back into DAS for the next reconstruction. Iteration cycle 1 repeats this sequence without reloading data, instead, the Deviation module first subtracts the previous image from the current weighted residual. Subsequent cycles mirror cycle 1 until the loss criterion is satisfied, at which point the Deviation module outputs the final image.

2) Data Reuse Method: In a PA imaging system, the transducer elements commonly arranged around the imaging target often exhibit a certain degree of symmetry. Examples include the linear arrays, semi-circular arrays, circular arrays, and even spherical arrays that we frequently use. This indicates that using popular symmetric array configurations can often yield better results, both at the signal and image levels. When implementing model-based algorithms in hardware, it is necessary to pre-store the delay data from the DAS algorithm, as well as the phase shift data and amplitude weighting data from the s-Wave algorithm. For instance, assuming 128 transducer elements and an ROI size of 128×128, the storage size required for the delay data (with a precision of 10 bits) is $128 \times 128 \times 128 \times 10$ bits $\approx$ 21 *Mb*. Similarly, the storage size for phase shift data (with a precision of 10 bits) is $128 \times 128 \times 128 \times 10$ bits ≈ 21 *Mb* and storage size for amplitude weighting data (with a precision of 8 bits) is $128 \times 128 \times 128 \times 8$ bits ≈ 17 *Mb*. Altogether, storing these parameters demands approximately 60 *Mb* of on-chip storage resources, which imposes a significant burden on the storage capacity of FPGA. For 3D PA imaging, this will become much more challenging, requiring several *Gb* storage resources.

Our proposed data reuse method significantly alleviates the storage burden. Consider the linear array arrangement in Fig. 5(a), where both the array elements and the ROI are symmetrical about the central axis. The first element on the left and the last element on the right are symmetrical. In addition, the line segments of the same color have equal lengths which means the equal distances. Therefore, the distances from the last element to all pixel positions in the ROI can be derived from the corresponding distances from the first element to all pixel positions due to this symmetry. In this way, we only need to store the parameters corresponding to the red array elements on the left side. Correspondingly, the circular array shown in Fig. 5(b) can be analyzed to exploit symmetry for data reuse. Taking a 6×6 imaging area surrounded by 24 sensors arranged in a circular array as an example, the two sensors indicated by the line segments of the same color are symmetric with respect to the origin point. This symmetry allows us to selectively store data for a subset of sensors, which is red as illustrated in Fig. 5(b), efficiently reducing storage requirements. Extrapolating this concept to 128 sensors which is also what we actually used in this paper, we only need to pre-store data corresponding to 17 sensors at a minimum, which is more than 7 times storage reduction. Considering the trade-off between area and speed, we decided to pre-store the parameters for 33 sets of sensors, which resulted in more than 3 times resource savings. This optimization allowed us to implement a 32-channel data parallel strategy within the FPGA, significantly enhancing the efficiency and performance. The implementation of the address multiplexing in hardware is accomplished by configuring 4 constraint-based counters, which also implies four addressing modes and a multiplexer in Fig. 5(c), following center-symmetry rules to construct an address mapping unit (AMU) for data access during various execution cycle (for a total of 128 channels of data, with 32 channels processed in parallel, $cc \leq 128/32 = 4$).

3) Hardware Implementation of DAS: In the DAS implementation circuit as shown in Fig. 6, we pre-store delays in 33 read-only memories (ROMs) for each sensor channel, with a delay data size of 128×128 to match the image size (the image grid size of the ROI region is set to 128×128). When the sensor data in the LU becomes available, it is fed

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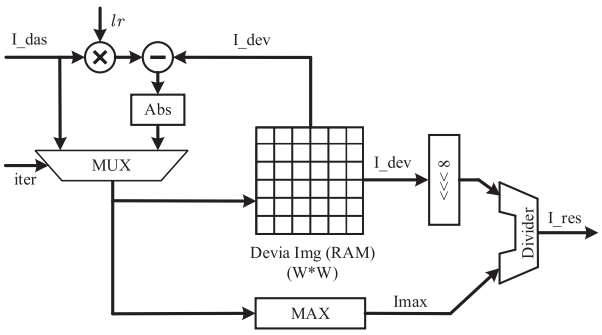


Fig. 7. Structure of the Deviation module. W*W: Image size.

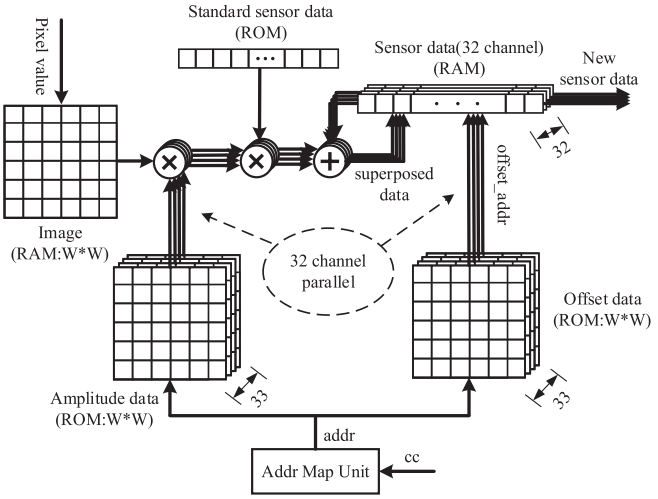


Fig. 8. Structure of the s-Wave module. W*W: Image size. cc: execution cycle count.

to the DAS module as soon as possible. To enhance data processing efficiency, we employ a 32-channel parallel data processing strategy, requiring 4 execution cycles to process the 128 channels of sensor data. Utilizing the data reuse method previously discussed, we enable 32 out of the 33 delay ROMs for each execution cycle. Specifically, when $cc = 1$ or 3, ROM0-ROM31 are enabled, and when $cc = 2$ or 4, ROM1-ROM32 are enabled. Additionally, the AMU unit adjusts the addressing mode of the delay ROM based on the execution cycle to ensure that the correct delay information is matched with the corresponding sensor channels.

Once the sensor data streams into the RAM, the DAS module begins addressing the ROM to retrieve the delay values. These delay values serve as addresses to fetch the corresponding delayed sensor values from the RAM. We employ a three-stage pipeline to sum the retrieved sensor values. The summed result is added to the corresponding position in the RAM, which stores the final image (initial value = 0). During the last execution cycle, the written image values are taken as absolute values and written back to the same positions. This accumulation process involves simultaneous read and write operations in the RAM, with the write data stream trailing the read data stream by two clock cycles.

To prevent unnecessary large data values caused by subsequent iterative algorithms, we can normalize the entire image, constraining the image values within the range of 0 to 256. The Max unit is responsible for finding the maximum value in the whole image while writing the image values during the final execution cycle. It consists mainly of a comparator, a selector, and a local register array (initial value = 0). After the DAS image accumulation is completed, the pipeline output is left-shifted by 8 bits to achieve a 256-fold magnification then fed into a divider along with the maximum value to perform normalization.

4) Hardware Implementation of Deviation Module: As illustrated in Fig. 7, the input of the Deviation module is an image generated from the DAS. Inside the module, the data flows along two paths. One path applies a learning rate *lr* weighting to the image, subtracts it from the previous iteration image stored in RAM, computes the absolute value, and then inputs it into a multiplexer. The other path directly inputs the image into another port of the multiplexer. The multiplexer selects its output based on the iteration count t of the model-based algorithm. When $t = 0$, the DAS image is directly output from the multiplexer and written into the RAM. For $t > 0$, the multiplexer outputs the residual between the two iterative images and writes it into the RAM, overwriting the previous image.

Once the residual image is written, it is output from the module as quickly as possible. During this process, the image range is constrained to 8 bits using circuitry similar to the output design of the DAS module. The direction of the output depends on the status of the iterative algorithm. If the iteration has not ceased, the output is fed into the s-Wave module for further processing. Otherwise, the Deviation module outputs the final reconstructed image. All the aforementioned processing is performed using a pipelined approach to increase the throughput of the module.

5) Hardware Implementation of s-Wave: The s-Wave module, as illustrated in Fig. 8, is designed with preloaded 33 sets of amplitude weighting data, 33 sets of offset data, and a single set of standard sensor data s stored in ROMs. Upon the complete transfer of pixel values from the Deviation module into the RAM, the s-Wave module initiates its processing sequence. Initially, pixel values are sequentially retrieved, simultaneously with the 32 amplitude values fetched from the ROMs. The enablement of 32 out of the 33 ROMs is determined by the current execution cycle. The pixel value is then broadcasted and multiplied by the 32 amplitude values, an operation that is efficiently completed within one clock cycle. These resultant values are then weighted and subsequently added to the standard sensor data, which has been broadcasted from the ROM, effectively modifying its amplitude characteristics. Next, the phase alteration is achieved by adjusting the start and end positions of data writing in the RAM. Specifically, the 32 phase offset data are read from the ROMs and utilized as read addresses to fetch the corresponding sensor data from the 32 RAMs. The amplitude-weighted standard sensor data are then summed with these retrieved sensor data corresponding to the offset data, and the results are stored back in their original RAM positions. During this accumulation phase, the generation of RAM write addresses is achieved by simply delaying the offset data by three clock cycles using a local register array. The ROM-stored amplitude and offset data are addressed correctly through the AMU, capable of generating four distinct addressing modes, analogous to those used in the DAS module. After all pixel values within the input image are processed, the s-Wave module completes the generation of 32 new sensor data sets, achieving full output

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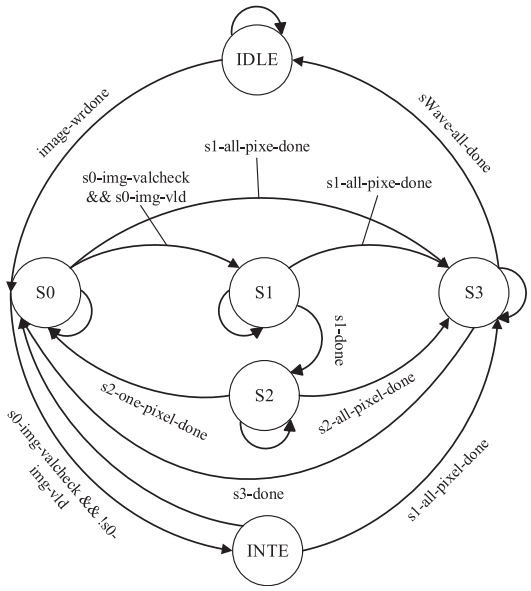


Fig. 9. State transition diagram of s-Wave algorithm hardware architecture design.

of all 128 channels of new sensor data within four execution cycles.

It is crucial to highlight that, for each individual pixel value, we must perform amplitude weighting across the entire length of the standard sensor data, followed by a systematic accumulation of data from RAM. This intricate processing sequence involves an unrolling procedure, necessitating the use of a finite state machine (FSM) to manage the conditional state transitions effectively. The state transition diagram of s-Wave is shown in Fig. 9, the finite state machine comprises six states: IDLE, S0, S1, S2, S3, and INTE, each orchestrating a distinct phase of the s-Wave processing pipeline. In the IDLE state, the system awaits the image-wrdone trigger before transitioning to S0, where pixels are read sequentially and validated. Valid pixels prompt a move to S1 for amplitude weighting, whereas invalid pixels redirect the machine to INTE, which immediately returns to S0 to fetch the next pixel. From S1, the machine advances to S2 to apply the requisite phase adjustment. Once all phase shifts are applied, execution enters S3, where the newly generated sensor data are output. State transitions are governed by explicit validity and completion signals, ensuring coherent interaction among states and enabling the module to complete its full processing cycle. By employing an FSM, we can meticulously orchestrate the various stages of the data processing pipeline, namely, the fetching, weighting, and accumulation of sensor data, ensuring that each operation adheres to both logical coherence and timing precision. Each state in the FSM corresponds to a specific phase of this pipeline, with state transitions dictated by predefined conditions based on execution cycles and data status. This method enhances the logical structure and timing efficiency of the data processing. The FSM framework provides granular control over the data processing flow, with each state representing a distinct phase of the pipeline.

6) Hardware Implementation of Loss Module: The Loss module, shown in Fig. 10, is tasked with computing the difference between the newly generated sensor data from the s-Wave module and the originally measured sensor data to

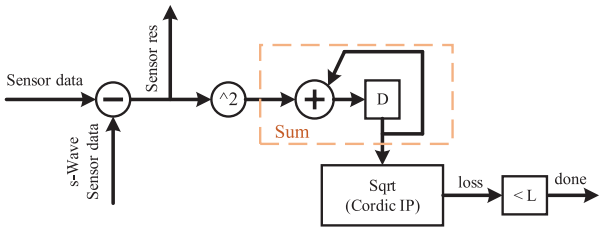


Fig. 10. Structure of the Loss module.

derive the sensor residuals, which are then squared and accumulated using an adder and a register to compute the sum of squares. Subsequently, the accumulated result is subjected to a square root operation using a CORDIC (Coordinate Rotation Digital Computer) algorithm IP core to obtain the final loss value. This loss value is then compared with a predefined threshold to determine whether the algorithm iteration should terminate, generating an iteration end signal based on this comparison.

III. Experimental Results

To demonstrate the feasibility of our proposed hardware architecture for accelerating the model-based algorithm with s-Wave and its potential integration into a palm-sized PAT system, we conducted experiments using both phantom and in vivo human finger data on an FPGA platform.

A. Experimental Setup

In our experiments, the PA signals were captured using a 128-element ring-shaped ultrasound transducer array with a radius of 30 mm [44], [45]. Data acquisition was handled by custom-designed data acquisition card (DAQ) capable of supporting 256-channel signal acquisition with a maximum sampling frequency of 80 MSPS and a maximum sampling depth of 4096 samples. The FPGA platform utilized is the Xilinx AXU15EG MPSOC, featuring a 200 MHz clock frequency generated from a differential clock signal supplied by an external crystal oscillator, 341280 Look-up tables (LUTs), and 26.2 Mb block RAM. As illustrated in Fig. 11, the experimental system is orchestrated such that the PC controls the signal generator to emit synchronized pulse signals to both the laser and the DAQ. The laser beam, directed through an optical fiber, targets the object located at the center of the ring transducer, generating PA signals that are subsequently received by the transducer. These signals are then transmitted to the DAQ, where they undergo amplification, filtering, and other preprocessing steps before being fed into the FPGA for image reconstruction. The reconstructed images are finally uploaded to the PC for display. The selected Region of Interest (ROI) (128×128 grids) measures 20 mm $\times$ 20 mm.

B. Implementation Results

1) Reconstructed Image Results: We obtained four sets of PA data by experiments: 1) cross-sectional images of three pencil leads placed perpendicular to the plane of the ring-shaped transducer; 2) cross-section images of two pencil leads placed parallel to the plane of the transducer; 3) in vivo human finger image; 4) and ex vivo breast tumor. Moreover, we also validated the FPGA reconstruction module using complex numerical vascular network phantom datasets. We employed

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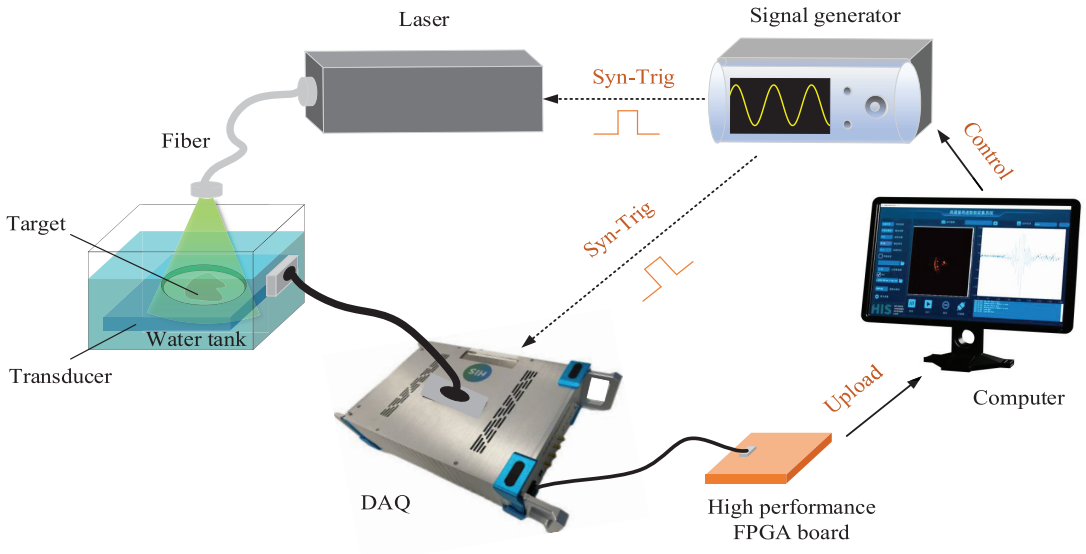


Fig. 11. The experimental setup of PA imaging system. DAQ: data acquisition card.

hardware implementations of the DAS algorithm, the DMAS algorithm, the DAS-CF algorithm, and our proposed model-based algorithm to obtain image reconstruction results for five sets of data, as illustrated in Fig. 12. Furthermore, we compared the results between the software-version model-based algorithm (s-Wave), hardware implementation and the original software-version model-based algorithm (k-Wave). The reconstructed images reveal that the DAS algorithm yields relatively coarse results, marred by significant noise and artifacts. However, both the DMAS and DAS-CF algorithms demonstrate substantial improvements over the DAS algorithm across all five datasets, as evidenced by their superior suppression of background noise, enhancement of the primary PA signal, and effective reduction of sidelobes. Compared to these DAS-based methods, the proposed novel hardware architecture of the iterative model-based algorithm not only effectively eliminates background noise and artifacts, but also significantly suppresses interference around and within the imaging targets. Additionally, it enhances the relative intensity of key pixels in the imaging targets, thereby greatly improving image contrast.

Detailed comparisons in the blue box, as illustrated in Fig. 12, show that in the reconstructed images of the three pencil leads, the contrast intensity of the two pencil lead sections within the white dashed box is noticeably lower in the DMAS and DAS-CF results compared to the model-based algorithm. This indicates a loss of some effective signals in the former two methods while suppressing sidelobes. For the results involving two pencil leads, although the DMAS and DAS-CF images exhibit a significant reduction in noise, a substantial amount of artifacts persists around the imaging targets, as shown by the white dashed box. In stark contrast, the results of our proposed method are virtually devoid of noise and artifacts. This trend is similarly observed in the results of finger, vascular network and breast tumor. It is particularly noteworthy that the DMAS-generated images of finger, vascular network and tumor images display numerous additional, albeit slightly weaker, pixels around some blood vessels, which can be attributed to spatial correlations among the image pixels. Furthermore, the DAS-CF algorithm exhibits marginally improved quality overall, particularly in the finger image, while the model-based algorithm are comparatively superior showing fewer interfering pixels around the blood vessels, thereby facilitating a clearer distinction between different vessels. Additionally, the model-based algorithm results, in contrast to those from the DMAS and DAS algorithms, enhance the contrast of specific pixels corresponding to the capillaries on the skin surface, thereby rendering the boundary of the finger and tumor more distinct. Notably, in vascular network images, although DMAS and DAS-CF suppress sidelobes effectively than DAS, they still leave substantial artifact interference, particularly between vessel branches, and attenuate some high-contrast vessels. Our model-based algorithm, by contrast, preserves vessel contrast while nearly eliminating both off-target and inter-vessel ghost signals. We can see that a low-intensity vessel near the boundary indicated by the second white dashed box, is partially retained by DMAS and DAS-CF, but is clearly resolved by k-Wave-based method. Our s-Wave-based method slightly diminishes its trailing edge and the hardware implementation further truncates it due to quantization. These observations suggest that while the proposed model-based algorithm (s-Wave) exhibits minor information loss for edge-located structures, it nonetheless preserves the critical features of the imaging targets.

To further investigate the differences between the results of the software-version model-based algorithm (s-Wave), hardware implemented model-based algorithm (s-Wave) and the results of the software-version model-based algorithm (k-Wave), we present the error maps relative to the k-Wave method results, as shown in Fig. 13. We can observe the error maps between the s-Wave-based and k-Wave-based software implementations exhibit an overall darker appearance, which indicates a high degree of similarity between the two results. In addition, even the lower intensity pixel clusters in the error maps continue to delineate the contours of the target objects, indicating the differences between the results of the two methods are largely confined to variations in pixel intensity. Moreover, closer inspection of the vascular error maps shows slight thickening of vessel contours, caused by minor differences between the s-Wave-based and k-Wave-based algorithms

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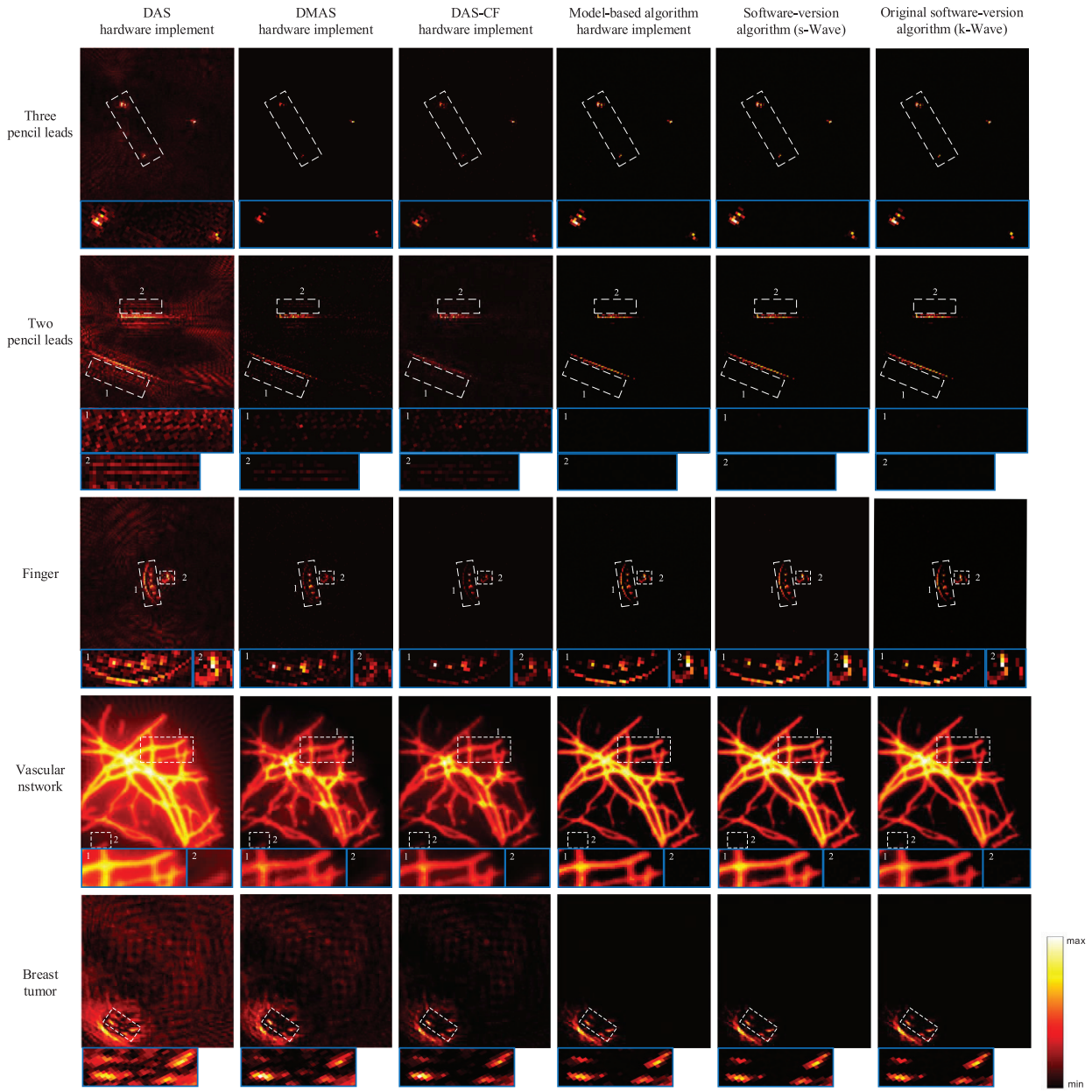


Fig. 12. Comparison of image reconstructed results from DAS (hardware implementation), DMAS (hardware implementation), DAS-CF (hardware implementation), the model-based algorithm (hardware implementation), the software-version model-based algorithm (s-Wave) and the original software-version model-based algorithm (k-Wave). The datasets used from top to bottom are: three pencil leads perpendicular to the transducer plane, two pencil leads parallel to the transducer plane, an in vivo human finger, vascular network and breast tumor. White dotted box: the chosen regions of interest (ROI). Blue box: zoomed ROI.

in handling target boundaries. This results in a small band of reduced-intensity pixels around the vessels as the second main discrepancy. However, in cases such as tumor, the error maps show almost no difference, demonstrating that s-Wave can closely replicate k-Wave performance under certain conditions. For the error maps of the hardware implementation and the k-Wave-based method, there are some extra differences, which are primarily due to the data quantization from floating-point to fixed-point within the FPGA. These observations suggest that the performance optimization achieved by the hardware implementation closely mirrors that of the original algorithm (k-Wave), which affirms the fidelity of the hardware implementation. Furthermore, it implies that the differences in pixel intensity can be effectively addressed through appropriate scaling adjustments, thereby ensuring the robustness and reliability of the hardware implementation.

We computed the structural similarity (SSIM) between the hardware-reconstructed images (DAS, DMAS, DAS-CF, hardware-implemented model-based (s-Wave), software-implemented model-based (s-Wave)) and those reconstructed by the original software-version model-based algorithm (k-Wave). As shown in Fig. 14, the SSIM values reveal that the DAS exhibits relatively poor performance, while the DMAS shows only average results. The DAS-CF algorithm performs reasonably well; however, its effectiveness markedly declines in the presence of substantial significant noise and artifacts. The SSIM values for the images of two pencil leads, three pencil leads, an in vivo human finger,

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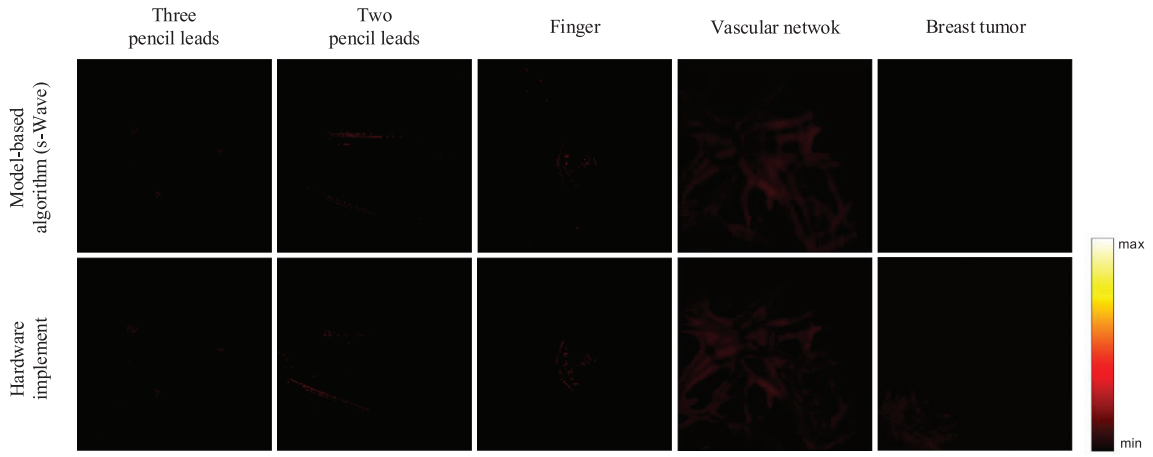


Fig. 13. The error maps between the results of the software-version model-based algorithm (s-Wave), hardware implementation and the original software-version model-based algorithm (k-Wave). These images are obtained by subtracting the results of the model-based algorithm (s-Wave) and hardware implementation from the original reconstruction results (k-Wave) and adding the absolute value.

TABLE I
PERFORMANCE COMPARISON

Platform	Intel i5 12500H				Xilinx AXU15EG	
Clock (MHz)	N/A				200	
Performance / Targets	Time (one frame)		Speed (FPS)		Time (one frame)	Speed (FPS)
	Float	int16	Float	int16		
Three pencil leads	2.054 s	1.682 s	0.487	0.595	0.0795 s	12.578
Two pencil leads	17.676 s	12.347 s	0.057	0.081	0.1332 s	7.508
Finger	12.715 s	8.862 s	0.079	0.113	0.0465 s	21.505
Vascular newtwork	168.25 s	112.312 s	0.0059	0.0089	1.1649 s	0.858
Breast tumor	23.687 s	15.853 s	0.042	0.063	0.6888 s	1.451

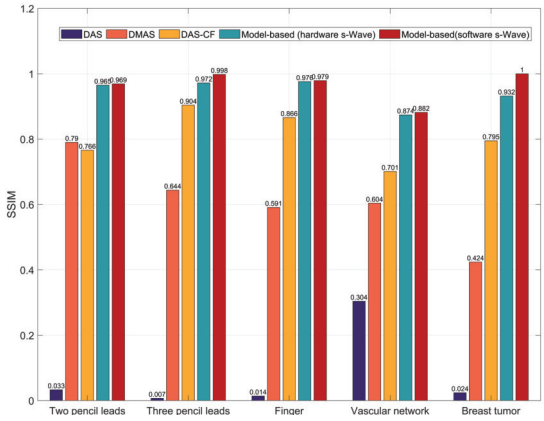


Fig. 14. SSIM of the results generated by the hardware implementation results of DAS, DMAS, DAS-CF and the Model-based algorithm (s-Wave), the software implementation results of model-based algorithm (s-Wave), comparing to the original software-version model-based algorithm (k-Wave).

vascular network and breast tumor reconstructed by the software-implemented model-based algorithm (s-Wave) and the original model-based algorithm (k-Wave) are 0.969, 0.998, 0.979, 0.882 and 1 respectively. It can be observed that the SSIM values of the software implementation are slightly higher than those of the hardware implementation, which can be attributed to minor errors introduced by data quantization in the hardware. Nevertheless, the average SSIM values for these two methods are 0.966 and 0.944, maintaining a consistently high level, which further corroborated the observation that the results obtained from the software-version model-based algorithm (s-Wave) and hardware implementation are exceedingly close to those produced by the original software-version model-based algorithm (k-Wave) and validated the reliability of the proposed model-based algorithm (s-Wave).

2) Hardware Implementation Performance: Given that the original model-based software algorithm utilizes floating-point arithmetic, which is prohibitively expensive for hardware implementation so that all algorithmic operations are employed by fixed-point quantization on the FPGA. To provide a more comprehensive comparison between software and hardware performance, we conducted experiments using both 16-bit integer precision and float precision in the software implementation. Since this paper represents the first implementation of a model-based algorithm on hardware, we compared the performance of our proposed hardware design with the software-version algorithm (s-Wave) running on a CPU, as detailed in TABLE I. The software-version algorithm (s-Wave) was developed using MATLAB R2021 and executed on a

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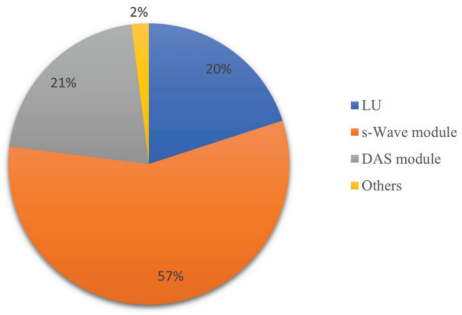


Fig. 15. Breakdown of the resource utilizations.

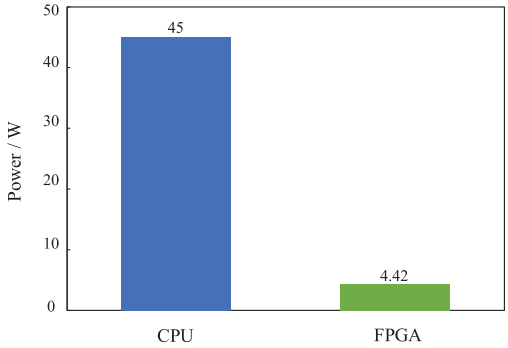


Fig. 16. Power comparison between CPU and FPGA.

TABLE II
RESOURCE UTILIZATION

Resource	LUT	LUTRAM	FF	BRAM	DSP
Used	145600	85587	58158	483	64
Available	341280	184320	682560	744	3528
Utilization	43%	47%	8.6%	65%	2%

computer equipped with an Intel i5 12500H CPU and 16 GB memory. The performance comparison results reveal that when the software operates using floating-point precision, while the CPU takes several seconds to tens of seconds to reconstruct one frame, the FPGA implementation achieves speeds approximately 26 to 270 times faster. When the software implementation is executed using 16-bit integer precision, the FPGA implementation still delivers a performance improvement on the order of approximately 21 to 190 times. Thus, it is evident that our proposed hardware-acceleration architecture delivers a significant speedup over a CPU in model-based reconstruction. In fact, our study focuses on accelerating the reconstruction process on the FPGA and does not delve deeply into front-end acquisition delays. However, we can also estimate the end-to-end frame time, including data acquisition, transmission, and reconstruction as follows. With 80 MSPS, 16-channel front-end DAQ, the acquisition time of one PA dataset (128×2048, where 128 is the number of channels and 2048 is the sampling depth we used) is $T_{acq} = \frac{128}{16} \times \frac{16 \times 2048}{80 \times 10^6} \approx 0.0033\,s$. Since only 1500 samples per channel are needed for FPGA reconstruction, the DAQ sends a reduced data block of $128 \times 1500 \times 16\,bits = 128 \times 1500 \times 2\,bytes \approx 0.366\,MB$ over a Gigabit Ethernet link (125 MB/s), the pure data-transfer time is $T_E = \frac{0.366\,MB}{125\,MB/s} \approx 0.0031\,s$. Thus, the front-end delay is $T_{DAQ} = T_{acq} + T_E = 0.0064\,s$, so the total capture time per frame is $T_{total} = T_{DAQ} + T_{recon}$, where T_{recon} is the FPGA image-reconstruction time from TABLE I.

The resource utilization of the FPGA (automatically calculated by Xilinx Vivado software) is detailed in TABLE II, with the corresponding allocation of major modules illustrated in Fig. 15. These numbers denote the number of units of each respective resource on the FPGA. A LUT is the FPGA's fundamental logic element, typically comprising a small RAM, a multiplexer, and a register, which is used to implement arbitrary Boolean functions. LUTRAMs are LUTs configured as low-latency, small-capacity RAM blocks, each providing tens to hundreds of bits (e.g., 64 bits) of storage. Each Flip-flop store one bit of state and, in conjunction with LUTs, enable sequential logic structures such as registers, counters, and finite-state machines. BRAM (Block RAM) is a dedicated large-capacity memory module in FPGA, which can be configured into a variety of bit width and depth combinations. The capacity of a single BRAM unit is usually 36 Kb (including check bits) or 18 Kb. DSP is a hardware module designed for high-performance mathematical operations, supporting multiplication, multiplication and addition (MAC), filtering and other operations. A DSP slice can achieve high-precision multiplication (such as 18×18 bits) or complex operations (such as complex multiplication). Specifically, the usage rates of LUTs, LUTRAMs, and BRAMs reached 43 %, 47 %, and 65 %, respectively, reflecting the logic- and memory-intensive nature of our 32-channel parallel processing architecture. As the fundamental logic units in FPGAs, LUTs are crucial for implementing complex logic circuits, and their high utilization highlights the heavy reliance of our computation-intensive tasks on logic resources. BRAMs supply large-capacity storage for parameters while LUTRAMs provide low-latency buffering for intermediate and image data, enabling high-throughput execution. FFs are heavily used for sequential logic, ensuring precise timing and data synchronization, whereas DSP utilization remains low since both DAS and s-Wave rely primarily on simple addition and multiply operations.

Additionally, we compared the power consumption of the CPU and FPGA, as shown in Fig. 16. The power consumption of the FPGA (automatically calculated by the built-in power analysis tool of Xilinx Vivado software) is more than 10 times lower than that of the CPU (automatically calculated by Intel Power Gadget software), resulting in an energy efficiency improvement of up to approximately 2700 times.

IV. CONCLUSION

In this paper, we present a novel implementation of a s-Wave-based iterative reconstruction algorithm for photoacoustic imaging on FPGA platform, achieving remarkable advancements in both speed and energy efficiency compared to traditional CPU implementations. The proposed hardware architecture utilizes an innovative data reuse method to optimize the use of storage resources, making it possible to be implemented in on-chip memory. Moreover, leveraging the s-Wave simplification of the photoacoustic forward model allows the hardware implementation of the forward and backward processing of the imaging algorithm, along with other operations, to be efficiently executed using simple multiplication, addition, and memory access operations. The

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implementation results show that the proposed FPGA accelerated architecture is capable of achieving a maximum approximately 21 fps, which underscores the feasibility of integrating the model-based algorithm into a palm-sized PAT system, paving the way for the development of real-time, compact, high-quality PA imaging devices.

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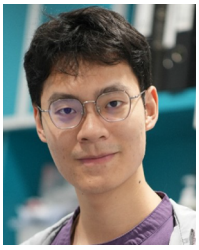
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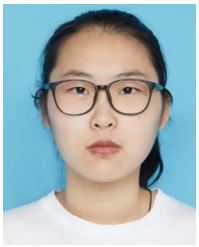
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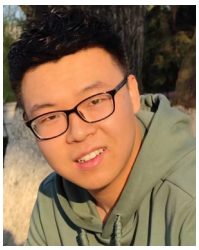
Yuwei Zheng received the B.S. degree from Harbin University of Science and Technology, Harbin, China, in 2022. He is currently pursuing the M.S. degree with the School of Information Science and Technology, ShanghaiTech University, Shanghai, China. His research interests include hardware acceleration of photoacoustic image reconstruction algorithms, low-cost and high-speed photoacoustic imaging system design, and its clinical application.



Zijian Gao received the B.S. degree from the Civil Aviation University of China, Tianjin, China, in 2020, and the M.S. degree from ShanghaiTech University, Shanghai, China, in 2023. He is currently pursuing the Ph.D. degree with the Faculty of Medicine, The Chinese University of Hong Kong, Hong Kong, China. His research interests include signal and image processing, digital circuit design, and quantitative biomedical imaging.



Yuting Shen received the bachelor's and master's degrees in computer science from ShanghaiTech University. Her research focuses on optimizing photoacoustic imaging reconstruction algorithms for heterogeneous media to enhance image quality. Her research includes developing photoacoustic non-line-of-sight algorithm and proposing the utilization of s-Wave for simulation acceleration.



Jiadong Zhang received the B.S. degree from Dalian University of Technology in 2020 and the M.S. degree from ShanghaiTech University in 2023. He is currently pursuing the Ph.D. degree with the Department of Biomedical Engineering, City University of Hong Kong. His research interests include computational imaging, artificial intelligence, and medical image analysis.



Ruixi Sun is currently pursuing the M.S. degree with the School of Information Science and Technology, ShanghaiTech University, Shanghai, China. His current research interests include FPGA-based algorithm acceleration and signal processing. His work involves developing efficient hardware architectures to enhance computational performance and optimizing signal processing techniques for high-speed data analysis.



Daohuai Jiang received the Ph.D. degree from the University of Chinese Academy of Sciences, Beijing, China, in 2023. In August 2023, he joined the Key Laboratory of Optoelectronic Science and Technology for Medicine, Ministry of Education, and Fujian Provincial Key Laboratory for Photonics Technology, Fujian Normal University. He is currently an Associate Professor at Fujian Normal University. His research focuses on the design and application of photoacoustic tomography systems. His primary interests include the development of novel photoacoustic imaging systems, biomedical signal processing, and their clinical applications.



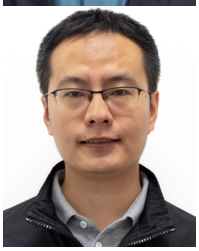
Fengyu Liu received the bachelor's and master's degrees from Xidian University in 2006 and 2009, respectively. He is currently an Engineer at the School of Information Science and Technology, ShanghaiTech University, where he is primarily responsible for the construction and management of the chip testing laboratory under the electronics discipline platform, while also providing technical support to researchers. He is also involved in the development of hardware circuits for research projects.



Xiran Cai (Member, IEEE) received the Ph.D. degree in physical acoustics from Sorbonne University, Paris, France, in 2018. From 2018 to 2020, he received post-doctoral training at the Ferrara Laboratory, Stanford University, Palo Alto, CA, USA. He is currently an Assistant Professor with the School of Information Science and Technology, ShanghaiTech University, Shanghai, China. His research interests include ultrasound imaging methods for diagnosis and therapy.



Feng Gao received the bachelor's degree from Xi'an University of Posts and Telecommunications in 2009 and the master's degree from Xidian University in 2012. His research interests include image processing and digital circuit design. In addition, he devotes his efforts to the development of underlying technology of photoacoustic imaging and promote the clinical application of photoacoustic technology.



Yuan Gao (Member, IEEE) received the B.Eng. and M.Eng. degrees from Xi'an Jiaotong University, Xi'an, China, in 2009 and 2012, respectively, and the Ph.D. degree from The Hong Kong University of Science and Technology (HKUST), Hong Kong, in 2017. From 2017 to 2019, he was a Post-Doctoral Fellow with HKUST. In November 2019, he joined the School of Microelectronics, Southern University of Science and Technology (SUSTech), Shenzhen, China, and currently as an Assistant Professor. He has authored more than 30 international journals and conferences. His current research interests include analog and power integrated circuit design. He is an ITPC Member of the IEEE International Solid-State Circuits Conference (ISSCC). He was a co-recipient of the Best Paper Award in the 2023 IEEE ISCAS and the Best Student Paper Nomination at the 2023 IEEE CICC.



Fei Gao (Member, IEEE) received the bachelor's degree from Xi'an Jiaotong University, and the Ph.D. degree from Nanyang Technological University. He was an Assistant Professor with ShanghaiTech University from 2017 to 2024. In November 2024, he joined the School of Biomedical Engineering, University of Science and Technology as a Professor, leading the Hybrid Imaging System Laboratory (www.hislab.cn). He has published over 180 journals and conference papers with more than 3800 citations. His interdisciplinary research topics include photoacoustic (PA) imaging physics (proposed non-line-of-sight PA imaging, passive PA effect, PA resonance effect and imaging, phase-domain PA sensing, pulsed-CW hybrid nonlinear PA imaging, and TRPA-TRUE focusing inside scattering medium), biomedical circuits and systems (proposed miniaturization methods of laser source and ultrasound sensors, delay-line based smart DAQ system, and hardware acceleration for PA imaging), algorithm and AI (proposed frameworks, such as Ki-GAN, AS-Net, Y-Net, EDA-Net, and DR2U-Net), and close collaboration with doctors to address unmet clinical needs (some prototypes are under clinical trials). He serves as a TPC Member for the IEEE Ultrasonics Symposium. He received Chinese Government Award for Outstanding Self-Financed Students Abroad (2014) and the Springer Thesis Award (2016). He is serving as an Associate Editor for several journals, including *Photoacoustics*, *Medical Physics*, *Ultrasound in Medicine and Biology*, and IEEE PHOTONICS JOURNAL.

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